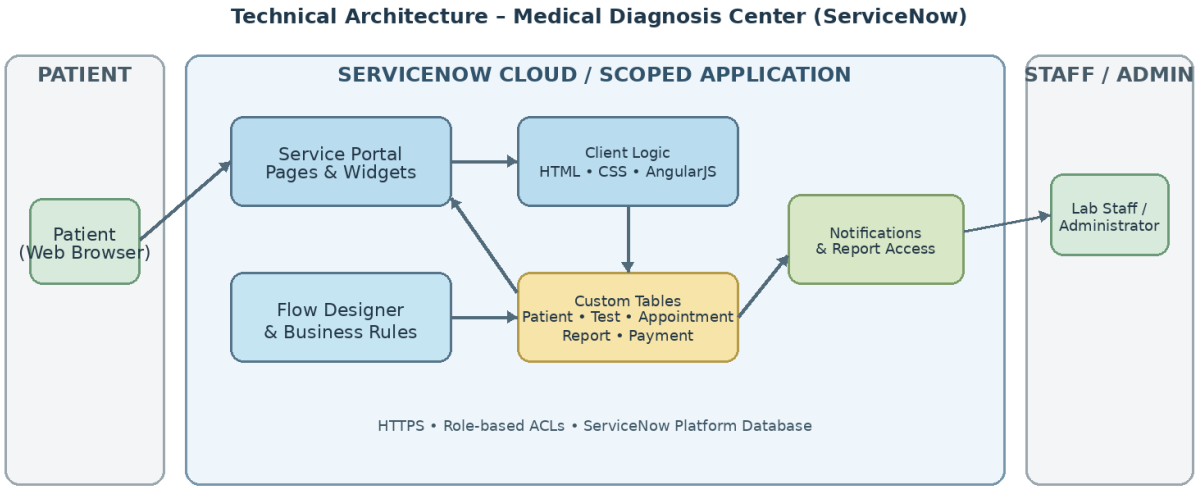
# Project Design Phase-II
# Technology Stack (Architecture & Stack)

| | |
|---|---|
| Date | 19 October 2026 |
| Team ID | |
| Project Name | Medical Diagnosis Center – Service Portal (ServiceNow) |
| Maximum Marks | 4 Marks |

## Technical Architecture:

The architecture below shows the patient-facing Service Portal, application logic, automation, custom data tables, notifications and staff access in the ServiceNow cloud. Details are summarized in Table-1 and Table-2.

**Example: Patient diagnostic test booking and report management through ServiceNow**




## Table-1 : Components & Technologies:

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | Patients use the portal to browse tests, book appointments, track status and access reports. | ServiceNow Service Portal, HTML, CSS, AngularJS |
| 2 | Portal Widgets | Custom pages/widgets support test selection, slot availability, booking summary and report access. | Service Portal widgets, HTML templates, CSS, client/server scripts |
| 3 | Client-Side Logic | Dynamic form behavior includes field visibility, price display and appointment-date validation. | AngularJS, Client Scripts, UI Policies |
| 4 | Application Logic | Server-side logic retrieves, inserts and updates records and validates booking data. | GlideRecord, Business Rules, Script Includes |
| 5 | Workflow Automation | Automates appointment submission, optional approvals, payment updates, cancellations and report notifications. | Flow Designer, Business Rules, Scheduled Jobs |
| 6 | Data Storage | Stores patient details, diagnostic tests, appointments, reports and payment transactions in related custom tables. | ServiceNow platform database, scoped application tables |
| 7 | Data Relationships | Reference fields connect patients to appointments, appointments to tests and reports, and related payment records. | ServiceNow reference fields, table relationships |
| 8 | Reports & Dashboards | Provides appointment and operational summaries for administrative monitoring. | ServiceNow Reports, Dashboards, charts |
| 9 | Notifications | Sends appointment confirmations, approval updates, reminders, payment updates and report-availability alerts. | Email Notifications, notification templates, event triggers |
| 10 | Security & Access | Restricts patient data and staff actions according to user roles and permissions. | ServiceNow authentication, roles, ACLs, HTTPS/TLS |
| 11 | Infrastructure | Application runs in the ServiceNow cloud development instance; no separate local server is required. | ServiceNow Personal Developer Instance (PDI) |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | Frameworks | Portal widgets use web technologies supported by ServiceNow Service Portal. | HTML, CSS, AngularJS |
| 2 | Security | Role-based access and ACLs control access to patient, appointment, payment and report records. | ServiceNow roles and ACLs, HTTPS/TLS |
| 3 | Scalability | Modular widgets, scoped tables and configurable flows support future feature additions. | ServiceNow scoped application, Flow Designer |
| 4 | Availability | The application is hosted on the ServiceNow cloud; development instances may hibernate after inactivity. | ServiceNow cloud / PDI |
| 5 | Performance | Client-side validation and platform-side queries/automation support responsive booking and record processing. | Service Portal, GlideRecord, platform caching |

## References:

https://www.servicenow.com/docs
https://developer.servicenow.com/
Project documentation: Medical Diagnosis Center Service Portal